

CONTIGUOUS MEMORY ALLOCATION

OPERATING SYSTEMS CONCEPT – LAB EXP – 09:

FIRST FIT

```
Activities Terminal Apr 20 17:07
snoordien@localhost:~/os-lab/firstfit — vim firstfit.c

1 #include <stdio.h>
2 int main(){
3     int i, n;
4     printf("Enter number of memory blocks:\n");
5     scanf("%d", &i);
6     int blockSize[i], remainingSize[i];
7     printf("Enter size of each block:\n");
8     for(int i=0; i<i; i++){
9         scanf("%d", &blockSize[i]);
10        remainingSize[i]=blockSize[i];
11    }
12    printf("Enter number of processes:\n");
13    scanf("%d", &i);
14    int processSize[i];
15    printf("Enter size of each process:\n");
16    for(int i=0; i<i; i++){
17        scanf("%d", &processSize[i]);
18    }
19    for(int i=0; i<i; i++){
20        int allocated = 0;
21        for(int i=0; k<k; k++){
22            if(remainingSize[k] >= processSize[i]){
23                int fragment = remainingSize[k] - processSize[i];
24                printf("Process %d of size %d is allocated to Block %d of size %d with fragment %d\n",
25                    i+1, processSize[i], k+1, blockSize[k], fragment);
26                remainingSize[k] = processSize[i];
27                allocated = 1;
28                break;
29            }
30        }
31        if(!allocated){
32            printf("Process %d of size %d is not allocated\n", i+1, processSize[i]);
33        }
34    }
35    return 0;
36 }
37
```

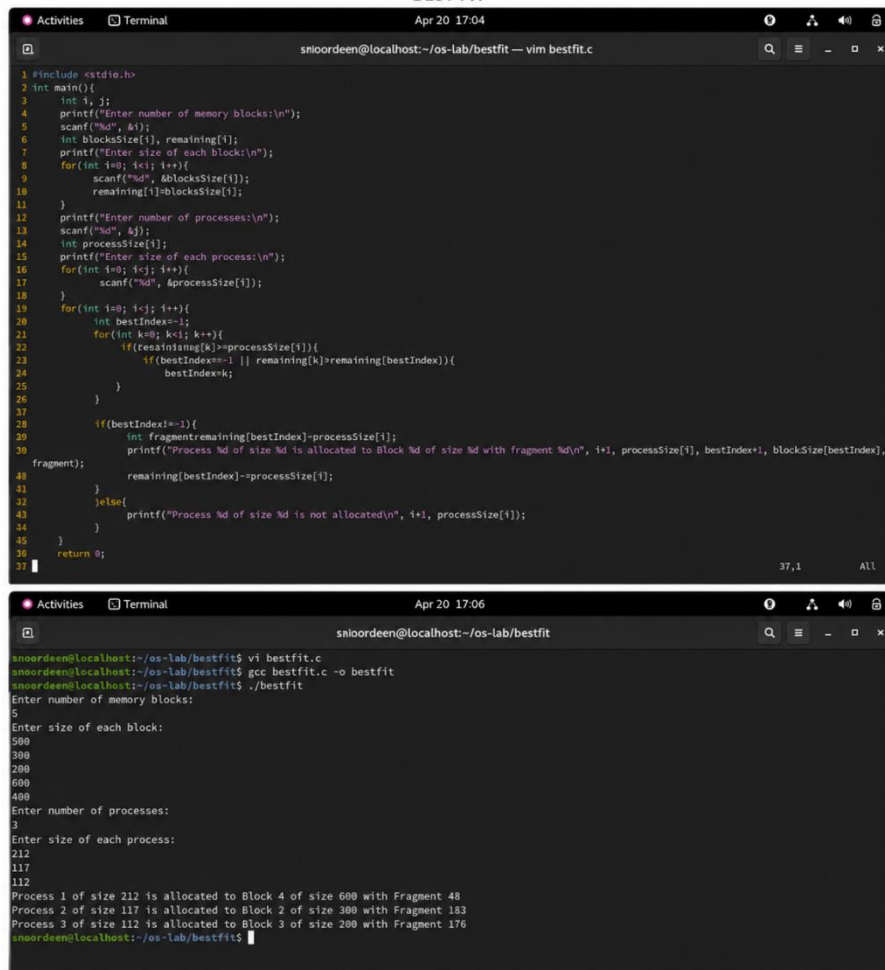
```
Activities Terminal Apr 20 17:09
snoordien@localhost:~/os-lab/firstfit

[snoordien@localhost firstfit]$ vi firstfit.c
[snoordien@localhost firstfit]$ gcc firstfit.c -o firstfit
[snoordien@localhost firstfit]$ ./firstfit
Enter number of memory blocks:
5
Enter size of each block:
500
300
200
600
400
Enter number of processes:
3
Enter size of each process:
212
417
112
Process 1 of size 212 is allocated to Block 2 of size 300 with Fragment 88
Process 2 of size 417 is allocated to Block 4 of size 600 with Fragment 183
Process 3 of size 112 is allocated to Block 2 of size 300 with Fragment 176
[snoordien@localhost firstfit]$
```

CONTIGUOUS MEMORY ALLOCATION

OPERATING SYSTEMS CONCEPT – LAB EXP – 09:

BEST FIT



```
1 #include <stdio.h>
2 int main() {
3     int i, j;
4     printf("Enter number of memory blocks:\n");
5     scanf("%d", &i);
6     int blockSize[i], remaining[i];
7     printf("Enter size of each block:\n");
8     for(int i=0; i<i; i++){
9         scanf("%d", &blockSize[i]);
10        remaining[i]=blockSize[i];
11    }
12    printf("Enter number of processes:\n");
13    scanf("%d", &j);
14    int processSize[j];
15    printf("Enter size of each process:\n");
16    for(int i=0; i<j; i++){
17        scanf("%d", &processSize[i]);
18    }
19    for(int i=0; i<j; i++){
20        int bestIndex=-1;
21        for(int k=0; k<i; k++){
22            if(remaining[k]>=processSize[i]){
23                if(bestIndex==-1 || remaining[k]>remaining[bestIndex]){
24                    bestIndex=k;
25                }
26            }
27        }
28        if(bestIndex!=-1){
29            int fragment=remaining[bestIndex]-processSize[i];
30            printf("Process %d of size %d is allocated to Block %d of size %d with fragment %d\n", i+1, processSize[i], bestIndex+1, blockSize[bestIndex],
31            fragment);
32            remaining[bestIndex]=processSize[i];
33        }
34        else{
35            printf("Process %d of size %d is not allocated\n", i+1, processSize[i]);
36        }
37    }
38    return 0;
39 }
```

```
snlordeen@localhost:~/os-lab/bestfit$ vi bestfit.c
snlordeen@localhost:~/os-lab/bestfit$ gcc bestfit.c -o bestfit
snlordeen@localhost:~/os-lab/bestfit$ ./bestfit
Enter number of memory blocks:
5
Enter size of each block:
500
300
200
600
400
Enter number of processes:
3
Enter size of each process:
212
117
112
Process 1 of size 212 is allocated to Block 4 of size 600 with Fragment 48
Process 2 of size 117 is allocated to Block 2 of size 300 with Fragment 183
Process 3 of size 112 is allocated to Block 3 of size 200 with Fragment 176
snlordeen@localhost:~/os-lab/bestfit$
```